ROCm Navigator
Autonomous CUDA-to-HIP Multi-Agent Migration Infrastructure

BUILT FOR BAND OF AGENTS & AMD INSTINCT™ HACKATHON 2026

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The Problem - GPU Code Migration Complexity & Friction

CODE GENERATION GAPS

Legacy tools like standard regular-expression hipify fail to
parse complex templates, cooperative group
synchronization primitives, and custom warp
configurations, producing invalid compile targets.

SDLC COORDINATION FAILURES

AI coding utilities operate in silos, failing to handle full
codebase compilation dependency order, verify
hardware-isolated build environments, or automatically
fix errors in loops.

THE REAL COST

Migrating CUDA codebases manually requires months of
specialist code rewrites and profile diagnostics, causing
corporate lock-in and high computing costs.

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Specialized Agents - A Digital Twin Engineering Team

PARSER & REWRITE AGENTS

Abdullah: Walks Clang AST cursor to extract global
functions, allocation calls, and templates. Builds NetworkX
topologies.
Malatesh: Manages asynchronous rewrite state matrices
using LangGraph and localized vector reference
databases.

VERIFICATION & PERFORMANCE

Arya: Validates synthesized code inside secure
Docker sandboxes and profiles runs via rocprof.
Yashwant: Intercepts leaks (Trufflehog), enforces
secure enclaves (TEE Vault), and syncs audit logs to
Neon PostgreSQL.

UI ORCHESTRATION & GATEWAY

Ansh: Handles integration layouts, FastAPI routing
gateways, WebSocket streaming logs, and the React Flow
dashboard canvas UI.

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End-to-End Autonomous SDLC Pipeline

STAGE 1: SECRETS
FILTER

Runs filesystem secrets audit
before LLM synthesis to prevent
credential leakage. Exits with 403
Halt if verified key leakage is
detected.

STAGE 2: AST
TOPOLOGY

Extracts call graph topology,
deadlocks, and warp metrics
using Clang AST mapping.

STAGE 3:
LANGGRAPH LOOP

Translates CUDA kernels to
HIP, injecting warp votes
suggestions and sync calls in
topological dependency order.

STAGE 4: BUILD
SANDBOX

Runs containerized compilation
under hipcc and captures
performance occupancy and
bandwidth metrics.

STAGE 5: RELEASE &
PR

Compiles PDF audit documents
and opens GitHub Pull Request
automatically.

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01. Repository Intelligence & AST Parser (Abdullah)

LIBCLANG AST EXTRACTOR

Use Libclang Python bindings (pip install libclang) to parse CUDA source files into a real
Clang AST. Walks the AST cursor to extract:
• Kernel declarations (CursorKind.FUNCTION_DECL with __global__ attribute).
• Memory allocation calls (CursorKind.CALL_EXPR matching cudaMalloc/cudaFree).
• Cross-file include chains and template instantiations.

NETWORKX TOPOLOGY GRAPH

Builds a NetworkX DiGraph from the extracted call relationships — nodes are kernel names, edges are launch-call dependencies. This processes and outputs a structured topology JSON including:
• call_topology: Adjacency map for compilation ordering.
• warnings: Pre-rewrite sync indicators.
• deadlocks_detected / difficulty_score: Metric arrays.
• warp_votes: Thread execution metrics.

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02. LangGraph AI Core Translation (Malatesh)

STATE GRAPH ORCHESTRATION

The LangGraph state must include the topology JSON object passed from the Architecture
Agent.
Before translating each file: read topology.warnings to inject CUDA-specific hints into the
system prompt (e.g. 'This file uses 32-thread warp votes — convert to 64-thread AMD
__ ballot').

SELF-HEALING COMPILER LOOPS

Uses topology.deadlocks_detected to conditionally insert hipStreamSynchronize calls
after multi-kernel sequences.
Uses difficulty_score > 60 as the condition to activate the self-healing compiler loop. The
call_topology adjacency map must be used to determine the correct order of kernel rewrites across files.

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03. DevSecOps Sandbox & Profiling (Arya)

ISOLATED COMPILES (hipcc)

Mounts synthesized HIP files inside a secure container using the rocm/dev-ubuntu-22.04
image.
• Spawns asynchronous Python controllers using the Docker SDK.
• Enforces strict 15-second compilation timeout thresholds.
• Prevents container privilege escalation bounds (privileged bounds strictly forbidden).

HARDWARE BENCHMARKS (rocprof)

Executes the test build inside sandboxes under the native AMD rocprof CLI tool.
• Traces hardware occupancy and memory bandwidth execution parameters.
• Computes efficiency bounds and flags modules under 0.85 occupancy ratings, routing
logs back to the self-healing loop.

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04. Security Audit & Neon PostgreSQL (Yashwant)

TRUFFLEHOG LEAK PREVENTION

Write a Python function using the subprocess module to execute a Trufflehog filesystem
scan (trufflehog filesystem <directory_path> --json) on the cloned repository temp
directory provided by the Gateway session store — this scan must run as the first step in
the pipeline, before any files are forwarded to the LLM Synthesis layer, so that no leaked
secrets or credentials ever reach the inference model. Parse the JSON output lines and format any detected credential leaks into a structured list of { detector: str, raw: str, verified: bool, filepath: str } dictionaries. If any verified leaks are found, emit a pipeline HALT signal back to the Gateway with a 403 status and a descriptive error message.

HIP STATIC ANALYSIS & VAULT

Additionally, build a POST /api/v1/security/scan endpoint that accepts { code: str, filename: str } for per-file static analysis of out-of-bound memory patterns on already-translated HIP code after the Rewrite Agent completes.
• TEE Vault: Secures key storage locally using Fernet symmetric encryption and TEE key
derivation (100k rounds PBKDF2 HMAC).
• Neon PostgreSQL: Implements SQLite-to-PostgreSQL SQL translation at runtime to
adapt and sync audit logs.

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05. Telemetry Interface & Handshake Gate (Ansh)

REACT FLOW TELMETRY CANVAS

• Renders real-time agent execution state transitions graphically via React Flow.
• Houses low-latency telemetry readouts (Overall Migration Score, Pipeline Progress, State
Tracker).
• Implements side-by-side git diff layout panels for reviewing source CUDA and target HIP
files.

WEBSOCKET STREAMING GATEWAY

• API Gateway exposes JWT-secured handshake handshake route `POST /api/v1/auth/session-handshake`.
• Establishes live WebSocket stream endpoint `GET /api/v1/dashboard/topology-stream` to
push asynchronous worker event logs directly to the browser UI with zero frame lag.

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Observable Event Bus & Collaboration

BAND SDK EVENT BUS

Rather than operating as a black box, ROCm Navigator leverages an event-driven model.
All agents publish operational telemetry to the shared event queue, including:
• ARCHITECT_SPEC_EMITTED: Structural specs are generated.
• FRONTEND_CODE_EMITTED / BACKEND_CODE_EMITTED: Source updates.
• CODE_REJECTED / CODE_APPROVED: Feedback logs.
• RELEASE_READY: Successful sign-offs.

REAL-TIME PIPELINE VISUALIZATION

Developers monitor multi-agent collaboration live via Band rooms. Every design decision,
security check, compiling trace, and self-healing adjustment is propagated synchronously
to the dashboard UI. This offers enterprise systems administrators full auditability and trace
logs for every modification made to proprietary compute kernels.

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ROCm Navigator - Complete Technology Stack

Layer Technologies Main Purpose
Frontend / UI Next.js 14, React Flow, WebSockets, Tailwind CSS Interactive telemetry and live diff canvas
Gateway & Bus FastAPI, Uvicorn, Band SDK Event Bus Token verification and real-time messaging
AST Parsing Libclang Python bindings, NetworkX Graph Engine Dependency trees and topology analysis
AI Translation LangGraph State Graphs, FAISS Vector Index Topological rewrites and self-healing loop
Sandboxed Build Docker SDK, hipcc, rocprof CLI tool Parity compilation and efficiency metrics
Data & Vault Neon PostgreSQL, cryptography (Fernet), SQLite Dialect adaptation, TEE encryption, audit logging

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"BUILD SAFER. SCALE FASTER. HEAL AUTOMATICALLY."

ROCm Navigator breaks hardware ecosystem lock-in, enabling zero-friction code mobility for enterprise GPU computing platforms.

Thank You! | AMD Instinct™ Hackathon 2026

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